

Contents of the lecture

- (1) Procedural Assignment
- (2) Blocking Assignment
- (3) Examples of Blocking Statements



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(1) There are two procedural blocks “initial” and “always”.

(2) Procedural assignments can only appear inside procedural blocks “initial” and “always”

(3) The LHS of procedural statement should be a register type (“reg”, “integer”, “real” and “time”)

(4) The RHS expression can be of net type or register type variables

(5) Two types of Procedural Assignment:

a) Blocking (denoted by “=”)

b) Non-Blocking (denoted by “<=”)

1. General syntax:
`variable_name = [delay_or_event_control] expression;`
2. The “=” is used to specify Blocking statement

3. The blocking assignment statements executes in the order they are written inside a procedural block
 - The target of the assignment is updated before the execution of next statement in the procedural block
 - They do not block the execution of statements in other procedural blocks

4. This is the recommended style for modeling combinational logic

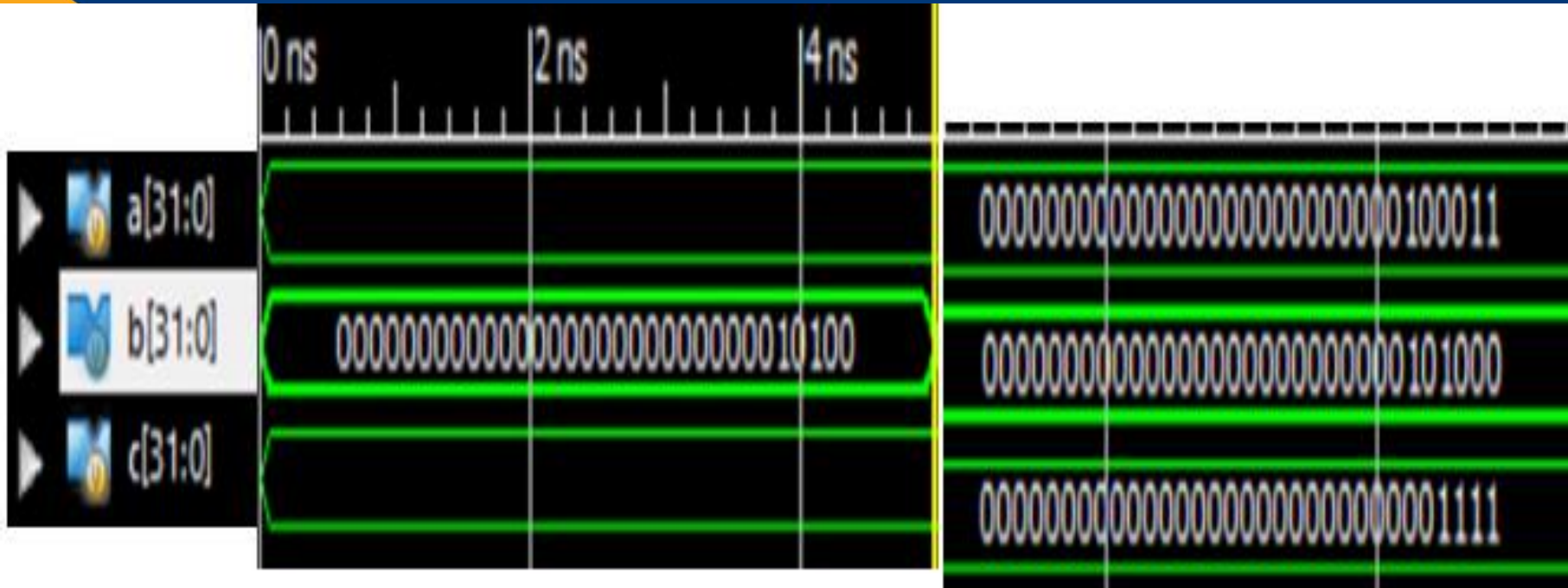
5. Blocking statements can also generates sequential circuit if all the values of case expression is not used in the program.

Results:
Initially, a=10, b=20, c=15
a becomes 35
b becomes 40
c becomes -5

[illegible]

Initially, a=10, b=20, c=15
a becomes 35
b becomes 40
c becomes 5

Blocking Assignment : Example-3



```
module ex( );  
integer a, b, c;  
initial  
begin  
a=10;  
b=20;  
c=15;  
a= b+c;  
b= #5 a+5;  
end  
endmodule
```

Results:

Initially, a=10, b=20, c=15
a becomes 35 at the start of simulation
b becomes 40 at 5 unit time of simulation

Contents of the lecture

- (1) Non-Blocking Assignment
- (2) Examples of Non-Blocking Assignment
- (3) Simulation result



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1. General syntax:
`variable_name <= [delay_or_event_control] expression;`
2. The “<=” is used to specify Blocking statement
3. The non blocking assignment statements execute concurrently
4. The value will be assigned to LHS only after the execution of all statements in a block
5. This is the recommended style for modeling sequential logic

Example-1

```
module ex( );  
integer a, b, c;  
initial  
begin  
a=10;  
b=20;  
c=15;  
end  
initial  
begin  
a<= b+c;  
b<= a+5;  
c<= a-b;  
end  
endmodule
```

Result:

Initially, a=10, b=20, c=15

a becomes 35

b becomes 15

c becomes -10

Example-2

```
module ex( );  
integer a, b, c;  
initial  
begin  
a=10;  
b=20;  
c=15;  
end  
initial  
begin  
a<= #5 b+c;  
b<= #5 a+5;  
c<= #5 a-b;  
end  
endmodule
```

All the RHS expression will be evaluated concurrently by using previous values of “a”, “b” and “c” and the evaluated values will be assigned to LHS together at 5 unit time.

Result:

Initially, a=10, b=20, c=15
a becomes 35 at 5 unit time
b becomes 15 at 5 unit time
c becomes -10 at 5 unit time